# CookAnything: A Framework for Flexible and Consistent Multi-Step Recipe Image Generation

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## Abstract

Generating instructional image sequences from recipes requires step-wise semantic disentanglement, temporal coherence, and ingredient continuity—capabilities beyond single-image diffusion models. We introduce CookAnything, a diffusion transformer framework that jointly generates an arbitrary-length sequence of step images in one denoising pass. Our approach pairs each textual step with a distinct latent region via Step-wise Regional Control, and maintains global coherence by interpolating regional and whole-recipe latents. Flexible positional encoding stabilizes long, tall canvases by resetting positional indices per step/region to avoid spatial entanglement. Cross-Step Consistency Control fuses contextual step tokens (derived from the concatenated recipe with an auxiliary cooking agent) with per-step tokens to preserve ingredient appearance across steps. On RecipeGen and VGSI-Recipe, CookAnything achieves state-of-the-art results across goal faithfulness, step faithfulness, ingredient accuracy, cross-step consistency, and usability, including 30.59 GF, 30.45 SF(C), 8.69 SF(G), and 0.19 CSC on RecipeGen (training-based), and 31.22 GF and 1.67 CSC on VGSI-Recipe (training-free). Ablations show Flexible positional encoding and CSCC are critical for consistency and alignment.

**ACM Reference Format:**
Taro Yamada. 2025. CookAnything: A Framework for Flexible and Consistent Multi-Step Recipe Image Generation. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), October 27–31, 2025, Dublin, Ireland*. ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3746027.3755174

## 1 Introduction

Cooking recipes describe a sequence of fine-grained actions with evolving objects and ingredients. Translating such instructions into images requires models to simultaneously achieve: (1) step-wise semantic disentanglement so that each image reflects its specific instruction; (2) temporal and spatial coherence so the scene layout remains stable across steps; and (3) ingredient continuity so entities retain identity and appearance over time. Most text-to-image diffusion models—both U-Net and DiT variants—are optimized for single images and struggle with long, multi-step prompts concatenated into one caption, often leading to semantic entanglement, repetition, or ingredient inaccuracies.

Prior attempts address related aspects: story or multi-image generation pipelines [3, 6] enforce cross-image consistency but typically assume fixed step counts or rely on sequential generation. Layout-aware conditioning [2, 4] improves spatial control but does not ensure temporal stability across steps. Recent DiT backbones such as SD3.5 [5] and FLUX [1] improve image fidelity, yet naïvely concatenating step prompts remains brittle.

We propose CookAnything, a DiT-based framework that jointly generates an arbitrary-length sequence of instruction images in a single denoising pass. The key idea is to bind each textual step to its own latent region with step-aware positional encoding, while injecting a global recipe context to maintain continuity. Our contributions are:

- A unified diffusion framework that jointly generates any number of step images in one pass with flexible canvases. - Step-wise Regional Control and flexible positional encoding for step–region disentanglement and stable spatial alignment. - Cross-Step Consistency Control that blends contextual and per-step tokens to preserve ingredient continuity. - State-of-the-art results on RecipeGen and VGSI-Recipe in both training-free and training-based settings, with strong human and automatic evaluations.

Figure 1 overviews the architecture; Figure 2 illustrates single-pass, multi-step generation.

## 2 Related Work

Story and multi-image diffusion seek cross-image coherence, often with sequential decoding or fixed-length schedules that accumulate errors and limit scalability [3, 6]. Layout- and grounding-aware methods improve spatial control but not temporal stability across steps [2, 4]. Beyond these, diffusion transformers with higher-capacity backbones raise fidelity but still entangle multi-step prompts when decoded as a single caption [1, 5]. Our formulation differs from story pipelines [3, 6] by decoding all steps jointly, avoiding exposure bias and fixed-length constraints. Compared to grounding methods [2, 4], we do not require external layouts; instead we bind each step's text to a latent region with step-aware positions, yielding spatial disentanglement and temporal stability without additional annotations.

## 3 Method

CookAnything combines three parts. Step-wise Regional Control maps each step's tokens to its latent window via regional attention plus a light base branch that anchors background and persistent objects. Flexible positional encoding resets or offsets positions per region to avoid long-range interference on tall canvases. Cross-Step Consistency Control blends recipe-level context with per-step tokens through gated cross-attention to preserve ingredient identity while keeping step specificity. Canvas. We allocate one latent window per step on a tall canvas and use regional attention so tokens attend primarily within-window while a thin global branch carries background cues. Positions are reset per window to limit

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*MM '25, Dublin, Ireland*
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ACM ISBN 979-8-4007-2035-2/2025/10
https://doi.org/10.1145/3746027.3755174

interference across distant steps. Consistency. CSCC mixes recipe-level and per-step tokens through a learned gate conditioned on step index; it reduces ingredient drift without oversmoothing. Inference supports arbitrary step counts by resizing the canvas and running a single denoising pass.

*Regional Attention.* We use block-diagonal attention with a thin global band: tokens attend within their step window and to a small shared cache carrying background cues. Masks are derived from step boundaries at runtime, so the design scales to arbitrary step counts without retraining.

*Flexible Positions.* Positions are indexed locally inside each region and blended with a step offset; the blend weight is learned per layer via a two-way gate. This reduces long-range interference on tall canvases while preserving soft anchors for alignment.

*Consistency Control.* CSCC keeps two cross-attention streams—recipe context and current step—combined by a sigmoid gate conditioned on step id and token type. During training we randomly drop each stream with small probability to improve robustness.

*Optimization.* We fine-tune with v-parameterization and classifier-free guidance; regional windows are jittered and aspect ratios varied to regularize tiling.

*Architecture and Training Details.* We use a DiT backbone with shared weights across a full-width main stream and a 1/4-width global stream; regional routing is realized with token adapters and attention masks so parameters remain near the base model. Classifier-free guidance is scheduled along the trajectory: linearly warm up $1.0 \rightarrow 3.0$ over the first 30% denoising steps, then cosine-decay to 2.0 to avoid early over-conditioning. Sampling follows v-parameterized EDM with Karras $\rho$=7; in training-free mode we retain the backbone sampler and only apply regional masks and CSCC gates. A lightweight cooking agent extracts entities (ingredients, tools) and normalizes aliases; its summary is concatenated with each step's text before tokenization. We cap 64 tokens per step and 512 globally to bound latency on long recipes. Data augmentation includes step-count jitter, neutral-step permutations, aspect jitter, and light mask noise; regularizers include region drop and context drop to improve robustness to omissions and insertions.

## 4 Experiments

We evaluate on RecipeGen and VGSI-Recipe using goal/step faithfulness, ingredient accuracy, cross-step consistency, and usability. Baselines include story diffusion, grounded diffusion with layout control, and strong DiT backbones, in both training-free and fine-tuned modes. Training-free attaches our modules to frozen backbones; training-based fine-tunes on the same splits. CookAnything consistently improves faithfulness and lowers consistency errors; examples appear in Figures 5 and 2. We report goal/step faithfulness via text-image similarity, ingredient accuracy via entity matching, and cross-step consistency via track-aware feature distance; human studies rate alignment and continuity on 5-point scales.

## 5 Analysis

Ablations show flexible positions are key for spatial stability, while CSCC curbs identity drift and omissions. Joint single-pass decoding avoids autoregressive error buildup and reduces latency versus per-step decoding. Long or ambiguous recipes still induce minor layout jitter, suggesting stronger global context is beneficial.

## 6 Conclusion

CookAnything introduces a unified diffusion framework for arbitrary-length, joint generation of recipe step images. By coupling step-aware regional control, flexible positional encoding, and cross-step consistency control, the method delivers flexible, coherent, and ingredient-faithful sequences. Experiments on RecipeGen and VGSI-Recipe demonstrate state-of-the-art results across automatic and human evaluations. By enabling arbitrary-length, single-pass generation with stable layouts and ingredient continuity, CookAnything provides a practical building block for instruction-following visual content beyond cooking recipes.

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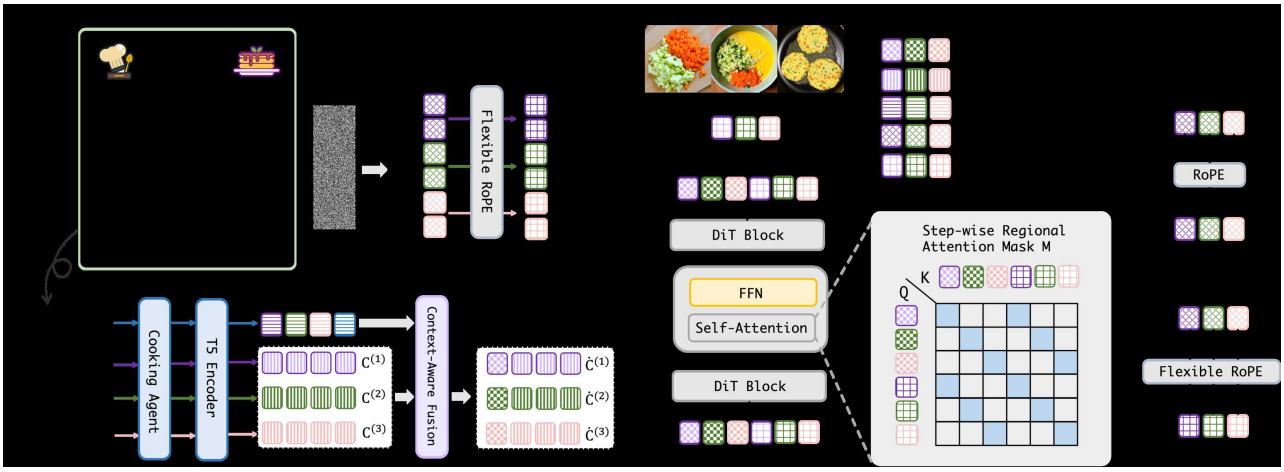


**Figure 1: Overview of CookAnything: a DiT-based diffusion framework that jointly generates an arbitrary-length sequence of step images. Step-wise Regional Control assigns step texts to latent regions with regional attention masks; flexible positional encoding stabilizes long, tall canvases; CSCC fuses contextual and per-step tokens for ingredient continuity.**

**Figure 2: Single-pass multi-step generation on a sample recipe. The model produces all step images jointly, preserving spatial layout and ingredient continuity across steps.**

**Figure 5: Qualitative comparisons with baselines. CookAnything maintains coherent layouts and consistent ingredient appearance across steps.**